FY27 Internship Assessment Package

Comprehensive Case Studies, Candidate Task Briefs, Synthetic Data Schemas, and Evaluation Rubrics

Program Overview & Assessment Strategy

This package provides four standardized case studies designed to evaluate finance, analytics, and software engineering candidates for the FY27 internship program. All four case studies are calibrated to a uniform difficulty level, requiring candidates to demonstrate end-to-end technical execution, handling of realistic data hygiene gaps, programmatic modelling, and strategic business synthesis.

Case Study Title	Primary Domain	Target Technical Capabilities
1. Install Base Monetization	Business Analytics / Strategy	SQL CTEs, Survival Analysis, Monte Carlo NPV, GTM Routing Logic
2. Data Science - Customer Lifetime Value Predictive Analytics	Data Science / Quantitative	Python ML (XGBoost/Survival), Back-Testing, Snowflake MLOps
3. Technical Data AI Validation	Full-Stack Software Eng.	React/Node.js, REST APIs, WebSockets, Snowflake Cost SQL
4. Environmental & Sustainability Specialist	Environmental & Sustainability Specialist	Advanced Excel (Pivot tables, GHG lookup tables, nested logic) or Python (pandas), ISO 14001:2026 Audit Standards, GRI Framework.

Case Study 3:
Technical Data AI Validation - Monetization (Pure Insights Command Centre)

- Domain: Full-Stack Software / Platform Engineering
- Calibrated Difficulty: Hard (4.0 / 5)
- Dataset File: `Case_Study_3_Pure_Insights_Monitoring.csv` (2,000 Rows)

Business Scenario & Problem Context

Following a migration from Tableau Server to a custom analytics portal (React, Node.js, Snowflake, AWS), the business lost native administrative visibility. The platform lacks centralized tracking for report usage, data refresh failures, report ownership, and Snowflake cloud compute costs. Candidates must build an operational full-stack admin dashboard with real-time WebSocket alerts and Snowflake cloud credit attribution.
Candidate Technical Mandate & Execution Tasks

- Task 1: Full-Stack Web Application: Develop a functional web dashboard using React (Frontend) and Node.js/Express (Backend) with Role-Based Access Control (RBAC) to monitor system health and orphaned reports.
- Task 2: Real-time Telemetry & WebSocket Alerting: Implement a server-sent events or WebSocket alerting system that triggers real-time operational notifications whenever dashboard load latency exceeds 15 seconds or automated refreshes fail.
- Task 3: Cloud Cost Allocation Engine: Write SQL queries against simulated Snowflake `ACCOUNT_USAGE` logs to attribute cloud credit consumption to individual reports, calculating unit cost per active user view.

Dataset Schema & Injected Data Anomalies (2,000 Records)

Candidates are expected to programmatically identify, clean, and document the following injected data quality defects:

- Orphaned report records missing owner email addresses (`NaN` or `'ORPHANED_REPORT'`).
- Negative Snowflake credit consumption values requiring anomaly filtering.
- Extreme load latency spikes (up to 55 seconds) requiring threshold alerting triggers.
- Discrepancies between active user counts and AWS hosting compute costs.

Required Candidate Submission Deliverables

1. Runnable Full-Stack Codebase (GitHub Repository with React frontend and Node.js backend).
2. REST API & WebSocket Technical Specification Document.
3. Snowflake Cloud Cost Allocation SQL Script Suite.

Internal Evaluator Grading & Assessment Rubric:

Full-Stack Web Architecture; Real-time Alerting & Telemetry: Cloud Cost SQL Attribution; Code Quality & System Docs